FuzzyCD: A Fuzzy Logic-Integrated Network for Robust Remote Sensing Change Detection With Noisy Labels

Jianlong Li, Ziyang Xiong, Xinxin Liu, *Member, IEEE*, and Bin Yang, *Senior Member, IEEE*

***Abstract*—Deep learning-based change detection (CD) has gained popularity in the field of remote sensing. However, most deep learning-based methods share a common limitation on the availability of massive high-quality labeled training data, where noisy labels are inevitably introduced during the annotation process. Noisy labels induce a surge in epistemic uncertainty and aleatoric uncertainty, thereby compromising the CD performance. Existing CD methods overlook the problem of noisy labels and fail to handle inherent uncertainties. To address this limitation, we propose a novel fuzzy logic-integrated network for robust CD with noisy labels, termed FuzzyCD, which integrates fuzzy logic to effectively model the two types of uncertainties (i.e., epistemic uncertainty and aleatoric uncertainty), mitigating the interference of noisy labels. Specifically, an epistemic uncertainty estimation (EUEU) unit is proposed, which treats XOR as fuzzy logic for feature fuzzification, thereby enhancing the model's cognitive capacity. In addition, an aleatoric uncertainty estimation unit (AUEU) is designed, which characterizes aleatoric uncertainty through probabilistic inference. By transforming the input feature space into a fuzzy feature space, the two fuzzy logic units significantly alleviate the negative impact of noisy labels, achieving improved CD performance and enhanced model robustness to noisy labels. Comprehensive experiments on three public CD datasets (e.g., LEVIR-CD, CDD, and WHU-CD) validate the effectiveness and superiority of the proposed FuzzyCD compared to other state-of-the-art methods.**

***Index Terms*—Change detection (CD), fuzzy logic, noisy labels, Remote sensing, uncertainty.**

I. Introduction

CHANGE detection (CD), as a pivotal technology in Earth observation, aims to capture landscape changes with remote sensing images acquired from the same geographical area but different periods [1], [2]. With the advancement of Earth observation satellites, CD has found widespread applications across diverse domains, including environmental monitoring [3], urban planning [4], deforestation monitoring [5], and disaster assessment [6], [7].

Received 30 July 2025; revised 20 January 2026; accepted 3 March 2026. Date of publication 10 March 2026; date of current version 19 March 2026. This work was supported in part by the Science and Technology Innovation Program of Hunan Province under Grant 2024RC3108; and in part by the Natural Science Foundation of Hunan Province, China under Grant 2023JJ20018. *(Corresponding author: Bin Yang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3672449

Deep learning offers an innovative technical solution for CD through automatically capturing latent data distributions and patterns, demonstrating significant performance improvements in recent studies [8], [9], [10]. Owing to the strong nonlinear modeling capabilities, convolutional neural networks (CNNs) [11], [12], [13] have achieved great success. For example, Zhao et al. [14] developed a novel CD method for complex scenarios based on feature interaction and multitask learning. Qu et al. [15] fused temporal and differential information for CD, successfully capturing potential correlations between bitemporal images. To better extract change representations, Feng et al. [16] designed a dual-branch multilevel intertemporal network, achieving improved CD performance. Notably, CNN models generally struggle to extract global contextual information due to their fixed receptive fields, while transformer-based methods have demonstrated superior capability in modeling long-range dependencies [17], [18]. For example, Chen et al. [19] introduced transformers into the CD task to better model contexts within the bitemporal images. An innovative CD network based on a transformer was proposed by Pan et al. [20] to effectively capture change semantics in small objects. Furthermore, some studies [21], [22], [23], [24], [25], [26] developed synergistic architectures that integrate the complementary strengths of CNN and transformer, which have gained popularity. Latest advances leveraging diffusion models [27], [28], [29], [30] and Mamba [31], [32], [33], [34] have significantly promoted the development of CD.

While the aforementioned deep learning-based CD methods have achieved remarkable performance, their effectiveness significantly deteriorates in real-world scenarios, where noisy labels are inevitably introduced due to the inherent data uncertainty and subjective manual annotations [35], [36]. Noisy labels, referring to incorrect annotations, distort the model's optimization trajectory. Fig. 1 illustrates two typical types of noisy labels prevalent in the field of CD, that is, unchanged areas erroneously labeled as changed, and changed areas erroneously labeled as unchanged, corresponding to case 1 and case 2 in the figure, respectively. Most existing CD methods overlook the problem of noisy labels, assuming that training labels are inherently clean. As training progresses, these methods eventually learn from the corrupted data distributions, leading to a degradation in CD performance.

To address the challenge of noisy labels, various methods have been developed and shown promising results, which

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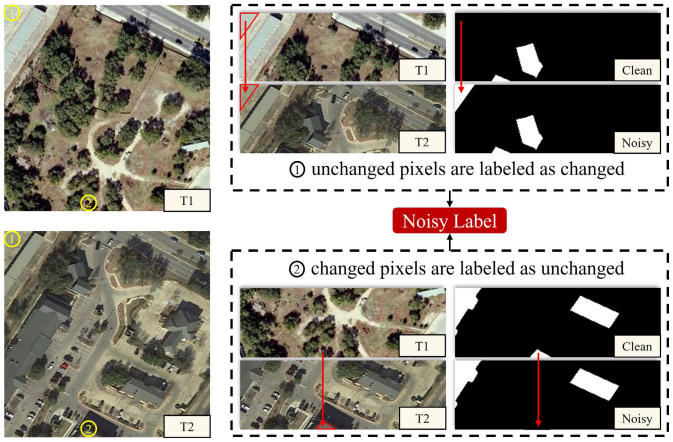


Fig. 1. Illustration of noisy labels. Case 1 and case 2 demonstrate two different noise types in CD.

are primarily applied to classification and segmentation tasks. According to their coping strategies, these methods can generally be categorized into three types: transition matrix-based methods [37], [38], [39], [40], sample selection-based methods [41], [42], [43], [44], and regularization-based methods [45], [46], [47]. The first category, transition matrix-based methods, tries to capture the transition probability between the noisy labels and true labels (i.e., ground truth, denoted as GT) [35]. The second category, sample selection-based methods, aims to differentiate clean samples and noisy samples, selectively discarding or correcting noisy instances to mitigate the impact of label noise [43]. The third category, regularization-based methods, employs a robust loss function or regularization term to enhance the model's robustness against noisy labels [45].

Despite the excellent performance achieved by these methods, their effectiveness may be constrained in CD tasks due to the complexity of cross-temporal semantic modeling. Moreover, they neglect the uncertainty issues induced by noisy labels, which is particularly important for accurate CD. Generally, there are two main types of uncertainties one can model [48]: epistemic uncertainty, which refers to the uncertainty in model parameters, demonstrates highly vulnerable to noisy labels and exhibits exponential growth under noisy conditions; and aleatoric uncertainty, which originates from inherent noise and variations in training data. Several studies have been proposed to address uncertainty in CD. For example, Li et al. [49] developed an uncertainty-aware network for building CD, effectively overcoming the uncertainty challenges. Shen et al. [50] proposed a semi-supervised network with parallel weak and strong branches, enabling quantification of aleatoric uncertainty. Both of them have demonstrated outstanding performance; however, their interplay with noisy labels remains unexplored. In the presence of noisy labels, the data distribution becomes distorted while the model's parameter optimization is continuously disturbed by incorrect label information. This significantly impairs the capability of CD model to capture change information, leading to a dramatic amplification of both epistemic uncertainty and aleatoric uncertainty. Therefore, there is an urgent need to find an effective way to model the epistemic uncertainty and aleatoric uncertainty caused by noisy labels, fully exploring the intrinsic relationship between uncertainty and noisy labels.

Fuzzy neural networks transform the input feature space into a fuzzy feature space through fuzzy logic, which has demonstrated superior performance in handling uncertain information [51], [52], [53]. Pal and Mitra [54] point out that the role of fuzzy logic embedded in neural networks is to effectively mimic uncertainty. By incorporating flexibility and tolerance to imprecision in the decision-making process, fuzzy logic can enhance the model's capability to model uncertainty and imprecision inherent in real-world data, thereby elevating the robustness and adaptability of the model. For instance, Gao et al. [51] proposed a fuzzy neural network for opinion score distribution prediction, highlighting the importance of uncertainty. Yang et al. [53] designed a fuzzy learning module to deal with the uncertainty in color and edge information in fine-grained lip image segmentation, achieving superior performance. To handle the inherent uncertainty and vagueness of information interaction across modalities, a novel deep fuzzy multiteacher distillation network is proposed by [55]. These successful practices motivate a natural idea that integrating fuzzy logic into CD model to effectively model the epistemic uncertainty and aleatoric uncertainty arising from noisy labels, mitigating the negative impact of incorrect label information.

To tackle the aforementioned challenges, a novel fuzzy logic-integrated network is proposed for robust CD with noisy labels, termed FuzzyCD. As illustrated in Fig. 2, the proposed network integrates fuzzy logic to effectively address the challenges of epistemic uncertainty and aleatoric uncertainty induced by noisy labels, significantly mitigating their adverse impact on CD performance. Building upon the well-established capability of fuzzy logic in handling uncertain information, two innovative fuzzy logic units are pioneered. Specifically, an epistemic uncertainty estimation unit (EUEU) is proposed, which leverages the intrinsic compatibility between XOR and CD, employing XOR as fuzzy logic for feature fuzzification. EUEU effectively captures the inherent epistemic uncertainty caused by noisy labels, thereby enhancing the model's cognitive capacity and discriminative power. For aleatoric uncertainty modeling, an aleatoric uncertainty estimation unit (AUEU) is introduced, which adopts a Gaussian membership function to compute pixel-wise degrees of belonging to fuzzy sets, capturing intrinsic uncertainty in raw data representations. Notably, AUEU operates independently of model-extracted features by taking concatenated bitemporal images as input, thus eliminating potential interference from model parameters and ensuring more robust aleatoric uncertainty estimation. By systematically incorporating fuzzy logic theory, the proposed FuzzyCD achieves improved CD performance as well as enhanced model robustness to noisy labels. In summary, the specific contributions are as follows:

1) A novel CD network is proposed to cope with noisy labels, named FuzzyCD. By integrating fuzzy logic into CD model, our proposed FuzzyCD effectively models epistemic uncertainty and aleatoric uncertainty caused by noisy labels, thereby alleviating the negative impact of incorrect annotations.

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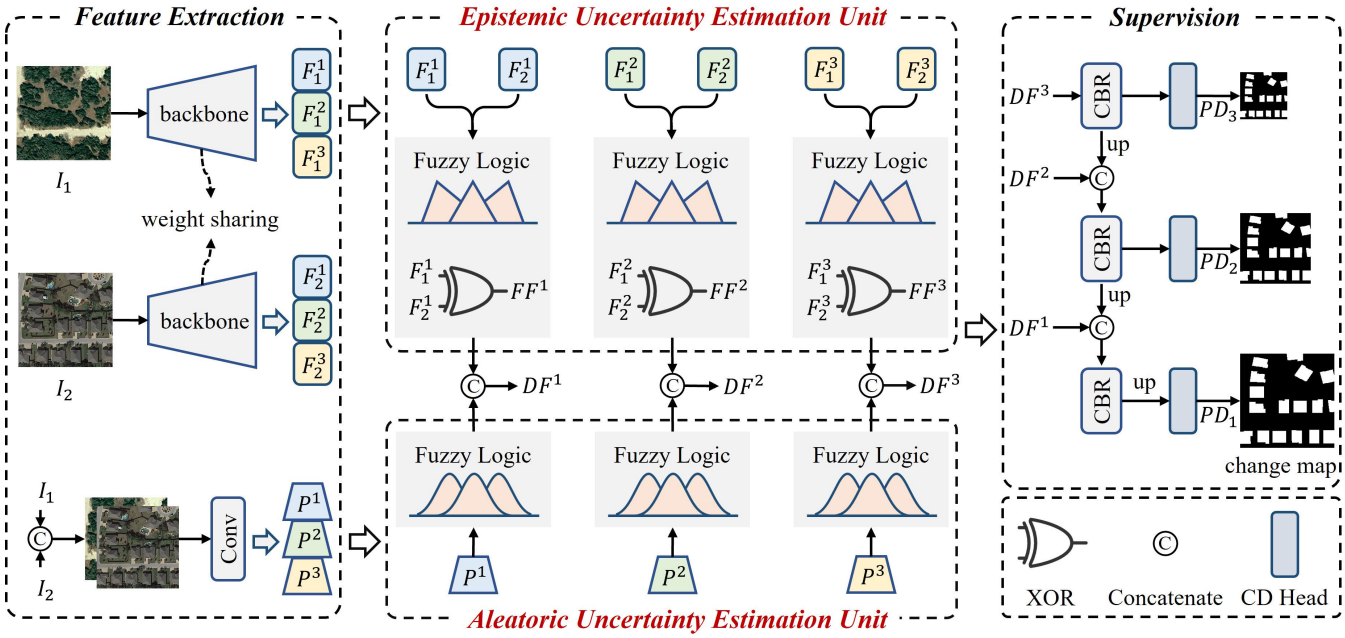


Fig. 2. Pipeline of the proposed FuzzyCD. The overall network consists of three key steps: feature extraction, uncertainty estimation, and supervision. The CBR represents a sequence of convolution, batch normalization, and ReLU.

2) The EUEU is proposed to effectively model epistemic uncertainty arising from noisy labels. By employing XOR as fuzzy logic for feature fuzzification, EUEU significantly enhances the model's cognitive capacity, thereby mitigating the interference of epistemic uncertainty and improving the CD performance.
3) The AUEU is pioneered to capture inherent aleatoric uncertainty. Through a specific membership function for probabilistic modeling, AUEU achieves better representation of complex data corruption, effectively reducing the aleatoric uncertainty and enhancing the model's robustness against noisy labels.

The rest of this article is structured as follows. Section II elaborates the details of the proposed FuzzyCD. A comprehensive experimental evaluation is presented in Section III. Finally, the conclusion is given in Section IV.

II. Proposed Method

A. Architecture Overview

The flowchart of our proposed FuzzyCD is illustrated in Fig. 2, which comprises three principal steps: feature extraction, uncertainty estimation, and supervision. The FuzzyCD is described specifically below.

First, given $\{T_1, T_2\} \in \mathbb{R}^{H\times W\times C}$ represents a pair of bitemporal images, where H, W, and C denote height, width, and channel, respectively. Recognizing that epistemic uncertainty is intrinsically linked to model parameters whereas aleatoric uncertainty arises from data variability, two complementary feature representations are extracted, for example, model-dependent features F_j^i, and data-dependent features P^i optimized to mitigate noise interference through explicit characterization of data-level stochasticity, where $i \in \{1, 2, 3\}$ is the different scale, and j denotes the temporal phase. The former is extracted by a Siamese weight-sharing backbone network (i.e., ResNet50 [56]), while the latter is derived via convolutional operations applied to concatenated bitemporal images. This dual-stream feature extraction mechanism enables joint modeling of both uncertainty types.

Then, the two different feature representations are, respectively, fed into the designed EUEU and AUEU for feature fuzzification, which refers to the mapping of image features to a membership plane [57]. To address uncertainties and enhance model robustness against noisy labels, fuzzy logic theory is incorporated in this article, which captures vague or imprecise information through linguistic variables and fuzzy sets, offering a robust network for modeling uncertain relationships [58]. Specifically, the two extracted features are considered as fuzzy elements. Let $\mathcal{X} = \{x_1, x_2, \ldots, x_N\}$ denote the input feature space, where $N = H \times W$ represents the number of pixel pairs and each feature x_m is assigned a membership degree for different fuzzy sets. EUEU and AUEU, respectively, employ the XOR operation and Gaussian membership function to calculate the extent to which an input feature belongs to a specific fuzzy set

$$\mathcal{FF}, \mathcal{P} = f_E\left(F_1^i, F_2^i\right), f_A\left(P^i\right) \tag{1}$$

where f_E and f_A represent EUEU and AUEU, respectively. $\mathcal{FF}$ and $\mathcal{P}$ stand for the fuzzy feature spaces derived from EUEU and AUEU, respectively. Similar to [59], the fuzzy sets are defined as follows:

$$S = \{NC, C\}. \tag{2}$$

NC denotes none-change. C denotes change. Following this, a series of fuzzy rules based on "IF-THEN" rules is applied

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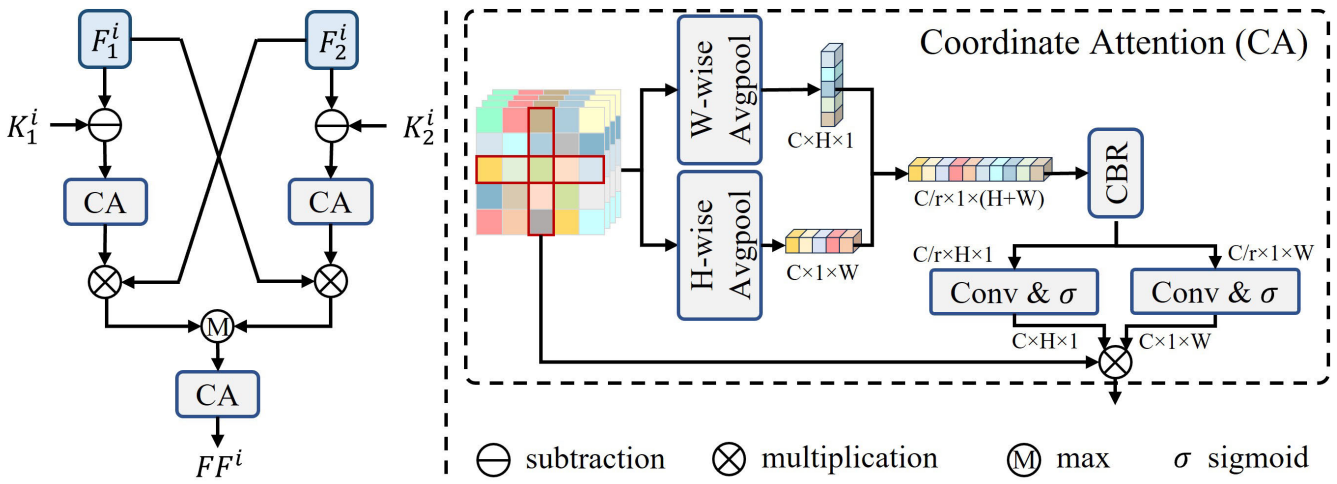


Fig. 3. Architecture of EUEU. The CBR represents a sequence of convolution, batch normalization, and ReLU.

for fuzzy inferencing

$$\begin{aligned} \text{Rule}_1 &: \text{IF } [\mathcal{FF}, \mathcal{P}] = NC, \text{THEN } \mathcal{O} = NC \\ \text{Rule}_2 &: \text{IF } [\mathcal{FF}, \mathcal{P}] = C, \text{THEN } \mathcal{O} = C \end{aligned} \tag{3}$$

where $\mathcal{O}$ represents fuzzy output set. $[\cdot]$ denotes the concatenation operation.

Finally, the multiscale fuzzy difference features DF^i are processed by CD prediction heads f_{head} under the deep supervision to obtain the final CD results PD_1, the process can be represented as follows:

$$PD_1 = f_{head}^1\left(f_{3\times3}^1\left(\left[f_{3\times3}^2\left(\left[f_{3\times3}^3\left(DF^3\right)\uparrow, DF^2\right]\right)\uparrow, DF^1\right]\right)\uparrow\right) \tag{4}$$

where $\uparrow$ denotes a bilinear upsampling operation and $f_{3\times3}^i$ represents a convolution layer with a filter size of 3×3, followed by batch normalization and nonlinear activation layers.

B. Epistemic Uncertainty Estimation Unit

Most previous deep learning-based CD methods overlook the critical role of uncertainty. However, in the presence of noisy labels, epistemic uncertainty is dramatically amplified. This can be explained by the fact that incorrect annotations will persistently corrupt the training process and adversely affect the optimization of model parameters. Consequently, noisy labels degrade the model's cognitive capacity, leading to an exponential growth in epistemic uncertainty. To tackle epistemic uncertainty, alleviating the interference of noisy labels with model cognition, as illustrated in Fig. 3, an EUEU is proposed to effectively model the epistemic uncertainty through the introduction of fuzzy logic, which has demonstrated strong robustness to uncertainty and noise.

The XOR operation is adopted by EUEU for feature fuzzification, which shares the same nature with CD. Specifically, if input bitemporal images are identical, the CD result remains unchanged (equivalent to XOR returning false); otherwise, the result is changed (equivalent to XOR returning true). Based on this, XOR is particularly well-suited for CD task, which can be expressed as follows:

$$A \oplus^L B = (A \wedge \neg B) \vee (\neg A \wedge B) \tag{5}$$

where $\oplus^L$ represents XOR operation. $\wedge$, $\vee$, and $\neg$ denote logical AND, logical OR, and logical NOT, respectively.

In the designed EUEU, the logical AND and logical OR can be approximated by multiplication and max, respectively. The logical NOT, while cannot be directly represented, can be derived from the observation that $\text{NOT}(x) = 1 - x$ for $x \in \{0, 1\}$. Similarly, trainable parameters K_j^i are introduced with an initial value of 1, allowing the logical NOT to be approximated as follows:

$$\neg F_j^i = K_j^i - F_j^i. \tag{6}$$

However, the conventional XOR operation is defined for binary inputs 0, 1, following the rules: $0 \oplus^L 0 = 0$, $0 \oplus^L 1 = 1$, $1 \oplus^L 0 = 1$, $1 \oplus^L 1 = 0$. To more exactly measure the degree of fuzziness, coordinate attention (CA) [60] is incorporated into EUEU, extending the fuzzy region to the interval [0, 1]. Furthermore, CA embeds positional information into channel attention, effectively capturing the long-range dependencies. As shown in the Fig. 3, CA mechanism comprises two steps: coordinate information embedding and attention weights generation. In the first step, channel-wise features are encoded along the horizontal and vertical coordinates, respectively. In the second step, directional attention weights are computed and applied to the original input. Mathematically, the process of EUEU can be formulated as

$$\begin{aligned} f_E\left(F_1^i, F_2^i\right) &= f_{CA}\left(\left(F_1^i \wedge \neg F_2^i\right) \vee \left(\neg F_1^i \wedge F_2^i\right)\right) \\ &= f_{CA}\left(\max\left\{\left(F_1^i \wedge \neg F_2^i\right), \left(\neg F_1^i \wedge F_2^i\right)\right\}\right) \end{aligned} \tag{7}$$

$$F_1^i \wedge \neg F_2^i = F_1^i \otimes f_{CA}\left(K_2^i - F_2^i\right) \tag{8}$$

$$\neg F_1^i \wedge F_2^i = f_{CA}\left(K_1^i - F_1^i\right) \otimes F_2^i \tag{9}$$

where f_{CA} represents CA. $\otimes$ denotes the pixel-wise multiplication operation.

C. Aleatoric Uncertainty Estimation Unit

Aleatoric uncertainty captures the inherent variability arising from noise in the data. Noisy labels, as a type of data corruption, particularly amplify this uncertainty, compromising the CD performance and model robustness. Critically, noisy

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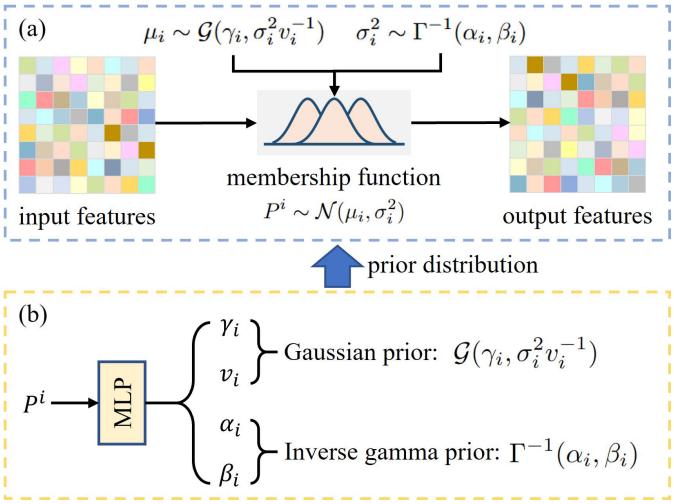


Fig. 4. Workflow of AUEU. (a) Feature fuzzification. (b) Parameter learning.

labels distort the optimization trajectory of the model, where epistemic uncertainty is propagated through erroneous gradient updates. This establishes a negative coupling whereby aleatoric uncertainty exacerbates epistemic uncertainty. To effectively characterize aleatoric uncertainty and further mitigate the negative impact of noisy labels, an AUEU is proposed in this article, which utilizes probabilistic inference to model the inherent data variability. The workflow of AUEU is visualized in Fig. 4.

To fully exploit the intrinsic data semantics with minimal model parameter influence, the bitemporal images are concatenated and processed through a convolutional layer for scale and channel transformation to maintain alignment with F_j^i. As indicated in Fig. 4(a), the P^i is assumed to follow a normal distribution with mean μ_i and variance σ_i^2:

$$\left(p_1^i, p_2^i, p_3^i, \ldots, p_N^i\right) \sim \mathcal{N}\left(\mu_i, \sigma_i^2\right) = \exp\left(-\frac{\left(p_m^i - \mu_i\right)^2}{2\sigma_i^2}\right) \tag{10}$$

where $\mathcal{N}(\cdot)$ denotes a Gaussian membership function, which establishes a mathematical bridge between precision and fuzziness by quantifying the degree to which features belong to fuzzy sets $\mathcal{S}$.

Notably, the mean μ_i and variance σ_i^2 are typically unknown in practice. Some methods [61] directly treat feature mean and variance as the parameters of a Gaussian membership function; however, it erroneously equates statistical parameters with semantic parameters, thereby compromising the model's capacity to characterize aleatoric uncertainty. The proposed AUEU implements adaptive parameter learning and iterative optimization through Bayesian estimation, achieving enhanced robustness against noisy labels.

Specifically, the μ_i and σ_i^2 are, respectively, placed with a Gaussian prior and an inverse gamma prior

$$\mu_i \sim \mathcal{G}\left(\gamma_i, \sigma_i^2 v_i^{-1}\right) = \sqrt{\frac{v_i}{2\pi\sigma_i^2}} \exp\left(-\frac{v_i\left(\mu_i - \gamma_i\right)^2}{\sigma_i^2}\right) \tag{11}$$

$$\sigma_i^2 \sim \Gamma^{-1}\left(\alpha_i, \beta_i\right) = \frac{\beta_i^{\alpha_i}}{\Gamma\left(\alpha_i\right)}\left(\frac{1}{\sigma_i^2}\right)^{(\alpha_i+1)} \exp\left(-\frac{\beta_i}{\sigma_i^2}\right) \tag{12}$$

where $\mathcal{G}(\cdot)$ represents Gaussian function. Γ represents the Gamma function. γ_i, v_i, α_i, and β_i are the learnable parameters of the Gaussian function and Gamma function. As shown in Fig. 4(b), a multilayer perception (MLP) is utilized to adaptively learn these parameters.

To further ensure effective parameter estimation and enhance model robustness against noisy labels, AUEU incorporates the negative log-likelihood loss to supervise the adaptive parameter learning process. Based on this, our objective is to estimate the posterior distribution $q(\mu_i, \sigma_i^2) = q(\mu_i, \sigma_i^2 | p_1^i, p_2^i, p_3^i, \ldots, p_N^i)$, which can be factorized as [62]

$$q\left(\mu_i, \sigma_i^2\right) = q\left(\mu_i\right) q\left(\sigma_i^2\right). \tag{13}$$

Since posterior distribution is generally intractable due to complex integration, the Gaussian conjugate prior (i.e., the normal inverse-Gamma distribution) is employed as an approximation. The specific form can be formulated as follows:

$$\begin{aligned} p\left(\mu_i, \sigma_i^2 | \gamma_i, v_i, \alpha_i, \beta_i\right) &= p\left(\mu_i | \gamma_i, \sigma_i^2 v^{-1}\right) p\left(\sigma_i^2 | \alpha_i, \beta_i\right) \\ &= \frac{\beta_i^{\alpha_i}\sqrt{v}}{\Gamma\left(\alpha_i\right)\sqrt{2\pi\sigma_i^2}}\left(\frac{1}{\sigma_i^2}\right)^{\alpha_i+1} \\ &\quad \times exp\left(-\frac{2\beta_i + v\left(\mu_i - \gamma_i\right)^2}{2\sigma_i^2}\right). \end{aligned} \tag{14}$$

Following this, the likelihood function and negative log-likelihood loss can be mathematically represented as (15) and (16), respectively,

$$\begin{aligned} &p\left(P^i | \gamma_i, v_i, \alpha_i, \beta_i\right) \\ &= \int p\left(P^i | \mu_i, \sigma_i^2\right) p\left(\mu_i, \sigma_i^2 | \gamma_i, v_i, \alpha_i, \beta_i\right) d\mu_i d\sigma_i^2 \end{aligned} \tag{15}$$

$$\begin{aligned} L_i^{NLL} &= -\log p\left(P^i | \gamma_i, v_i, \alpha_i, \beta_i\right) \\ &= -\sum_{m=1}^{N} \log p\left(p_m^i | \gamma_i, v_i, \alpha_i, \beta_i\right) = \frac{1}{2}\log\left(\frac{\pi}{v_i}\right) \\ &\quad - \alpha_i \log\left(2\beta_i\left(1 + v_i\right)\right) + \log\left(\frac{\Gamma\left(\alpha_i\right)}{\Gamma\left(\alpha_i + \frac{1}{2}\right)}\right) \\ &\quad + \left(\alpha_i + \frac{1}{2}\right)\log\left(v_i\left(p_m^i - \gamma_i\right)^2 + 2\beta_i\left(1 + v_i\right)\right). \end{aligned} \tag{16}$$

D. Noise-Robust Loss Function

Loss function serves as a fundamental component in deep learning-based supervised learning methods, critically influencing the model optimization and performance. In this article, the end-to-end training strategy is employed to optimize our proposed FuzzyCD. Specifically, three predictions PD_i with different scales are obtained through successive upsampling and concatenation operations, which are supervised through cross-entropy (CE) loss. The total loss is defined as follows:

$$L_{total} = \sum_{i=1}^{3}\left(L_i^{CE} + \lambda L_i^{NLL}\right) \tag{17}$$

where λ is the weighting factor of the negative log-likelihood loss L_i^{NLL}, balancing the contribution of the two losses, which

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will be discussed in the subsequent ablation experiments. L_i^{CE} stands for the CE loss

$$L_i^{CE} = -\frac{1}{N}\sum_{n=1}^{N}\left[Y_i^n \log PD_i^n + \left(1 - Y_i^n\right)\log\left(1 - PD_i^n\right)\right] \quad (18)$$

where Y_i denotes the noisy label at the ith scale, and n represents the pixel index. For the nth pixel, $Y_i^n = 0$ indicates nonchange and $Y_i^n = 1$ indicates change, with potential annotation errors.

III. Experiments

A. Datasets

To verify the effectiveness of the proposed FuzzyCD, extensive experiments are conducted on three widely used CD datasets.

1) LEVIR-CD [63]: LEVIR-CD is a public, large-scale building CD dataset released by Beihang University, capturing land-use changes over periods ranging from 5 to 14 years. It consists of 637 very-high-resolution (0.5 m/pixel) image pairs with a spatial size of 1024×1024. Considering the limitation of GPU memory capacity, each image is cropped into 16 nonoverlapping patches of 256×256 pixels, and they were randomly split into 7120/1024/2048 for training, validation, and testing, respectively.

2) CDD [64]: CDD is a season-varying dataset acquired from Google Earth, with spatial resolutions ranging from 0.03 to 1 m/pixel. It consists of 16 000 image pairs with a spatial size of 256×256, typically obtained from satellites or aerial photography to cover a diverse geographical area and changes at different periods. In this work, the image dataset was randomly divided into 10 000/3000/3000 for training/validation/test sets.

3) WHU-CD [65]: WHU-CD comprises one pair of bitemporal aerial photographs with a spatial size of $32\,507 \times 15\,354$ pixels and a resolution of 0.2 m/pixel. The original images are cropped into 256×256 patches with nonoverlapping, which are subsequently divided into training, validation, and test sets according to a ratio of 8:1:1, with a size of 6096/762/762.

All three CD datasets mentioned above are characterized by noise-free label annotations. To better simulate manual annotation errors, BIT [19] is employed to generate noisy labels. Specifically, the BIT network is trained on the three datasets, and its predictions on both training and validation sets are adopted as noisy labels in this article. The erroneous predictions indicate more challenging areas, that is, regions more likely to be annotated incorrectly, thus closely approximating real-world scenarios. To ensure sufficient noise levels, the hyperparameters configured in the original paper are intentionally adjusted, with the number of epochs set to 100 (compared to 200 in the original paper), thereby compromising the performance of BIT to introduce more noisy labels. Moreover, to investigate the robustness of the proposed FuzzyCD under different noise ratios, experiments are conducted with three noise ratios: $\rho \in \{0, 0.5, 1\}$, where $\rho = 0.5$ indicates that 50% of the label images are sourced from the prediction results of BIT, while the remaining 50% are sourced from the ground truth labels. Fig. 5 illustrates several sample images from the aforementioned three CD datasets along with their corresponding labels. As shown in the figure, the noisy labels exhibit blurred boundaries, with originally distinct objects becoming interconnected. Moreover, the majority of noisy labels concentrate in edge regions, which realistically captures the annotation ambiguity and uncertainty inherent in manual labeling, particularly for edge definition in CD tasks.

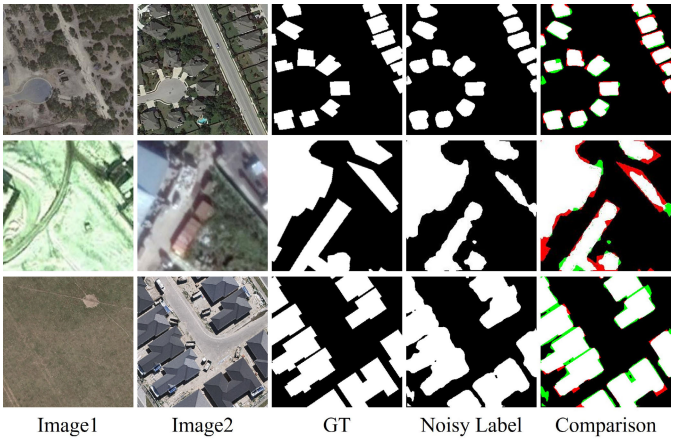


Fig. 5. Sample images from three datasets. The pre- and postremote sensing images are shown in the first two columns. The third and fourth columns illustrate the GT labels and noisy labels, respectively. The fifth column presents a comparative diagram between GT and noisy label, where red indicates false negatives (changed pixels mislabeled as unchanged) and green indicates false positives (unchanged pixels mislabeled as changed).

B. Comparison Methods

To authenticate the superiority of FuzzyCD, the proposed method is compared with nine state-of-the-art methods, including ICIFNet [18], MSCANet [66], DMINet [16], AMTNet [67], DA-IDANet [24], STFF-GA [68], SEIFNet [69], ISDANet [70], and EFICNN [8]. These approaches have attained competitive performance on CD task through different technical approaches. To ensure the fairness of the experiments, their hyperparameters are configured according to the settings specified in their original literature.

1) ICIFNet: An intrascale cross-interaction and interscale feature fusion network, which explicitly taps the potential of integrating CNN and transformer.

2) MSCANet: A CNN-transformer network with multiscale context aggregation, which combines the merits of CNN and Transformer to fulfill efficient and effective cropland CD.

3) DMINet: A dual-branch multilevel intertemporal network to efficiently derive the change representations by unifying self-attention and cross-attention in a single module.

4) AMTNet: An attention-based multiscale CNN-transformer network that combines the benefits of ConvNets, transformers, multiscale, and attention mechanisms.

5) DA-IDANet: A domain adaptive and interactive differential attention network, facilitating effective coupling of feature information.

6) STFF-GA: A CD network via spatio-temporal feature fusion and guide aggregation significantly reduces the impact of pseudo-changes and other interference factors.

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TABLE I
COMPARISON RESULTS ON THREE CD DATASETS. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUES (%)

Datasets	Methods	Year	$\rho = 0$	$\rho = 0.5$	$\rho = 1$
			Precision/Recall/F1/IoU/OA/Kappa	Precision/Recall/F1/IoU/OA/Kappa	Precision/Recall/F1/IoU/OA/Kappa
LEVIR-CD	ICIFNet	2022	92.02/86.31/89.08/80.30/98.92/88.51	93.28/82.46/87.54/77.84/98.80/86.92	92.85/80.61/86.30/75.90/98.70/85.62
	MSCANet	2022	90.66/88.15/89.39/80.81/98.93/88.82	92.22/84.43/88.15/78.82/98.84/87.55	93.09/80.68/86.44/76.12/98.71/85.77
	DMINet	2023	91.47/88.97/90.20/82.15/99.02/89.68	93.64/83.52/88.29/79.04/98.87/87.70	93.62/79.50/85.99/75.42/98.68/85.30
	AMTNet	2023	91.13/88.05/89.56/81.10/98.95/89.01	91.92/84.87/88.25/78.98/98.85/87.65	92.54/81.89/86.89/76.81/98.74/86.23
	DA-IDANet	2024	90.32/**89.98**/90.15/82.06/99.00/89.62	92.11/85.59/88.73/79.74/98.89/88.15	92.38/82.68/87.26/77.40/98.77/86.61
	STFF-GA	2024	90.25/88.93/89.59/80.77/98.93/88.80	91.03/85.36/88.10/78.74/98.83/87.49	92.09/80.77/86.06/75.53/98.67/85.36
	SEIFNet	2024	92.76/88.86/90.77/83.10/99.08/90.28	93.66/84.86/89.04/80.25/98.94/88.49	**94.63**/78.76/85.97/75.39/98.69/85.29
	ISDANet	2025	89.21/89.62/89.41/80.85/98.92/88.84	92.47/84.44/88.27/79.01/98.86/87.67	93.92/79.84/86.31/75.92/98.71/85.64
	EFICNN	2025	92.33/89.70/91.02/84.04/99.10/90.37	**94.57**/84.53/89.27/80.62/98.96/88.73	94.06/79.60/86.23/75.79/98.70/85.56
	Ours	-	**93.03**/89.73/**91.35/84.08/99.13/90.90**	94.10/**86.68/90.23/82.21/99.04/89.73**	94.37/**84.87/89.37/80.78/98.97/88.83**
CDD	ICIFNet	2022	92.30/87.65/89.91/81.68/97.68/88.61	93.45/80.68/86.59/76.36/97.05/84.95	95.50/77.92/85.82/75.16/96.96/84.14
	MSCANet	2022	93.40/89.57/91.45/84.24/98.02/90.33	95.82/82.90/88.89/80.00/97.56/87.52	96.36/79.61/87.18/77.28/97.24/85.65
	DMINet	2023	95.15/92.84/93.98/88.65/98.60/93.19	96.49/86.82/91.40/84.17/98.07/90.32	97.02/81.29/88.46/79.31/97.50/87.07
	AMTNet	2023	94.27/92.06/93.16/87.19/98.40/92.25	95.06/85.63/90.10/81.99/97.78/88.86	96.88/80.63/88.01/78.59/97.41/86.57
	DA-IDANet	2024	95.24/87.52/91.22/83.85/98.01/90.10	94.84/86.26/90.35/82.39/97.83/89.12	96.43/80.08/87.50/77.77/97.30/86.00
	STFF-GA	2024	94.04/90.14/92.09/83.97/98.02/90.17	92.42/87.11/89.69/81.30/97.64/88.35	92.69/83.89/88.07/78.69/97.32/86.57
	SEIFNet	2024	95.35/**94.92**/95.13/90.72/98.85/94.49	**96.94**/87.14/91.78/84.81/98.16/90.75	**97.57**/81.58/88.86/79.96/97.59/87.52
	ISDANet	2025	91.88/90.93/91.40/84.17/97.98/90.26	87.02/**92.93**/89.88/81.62/97.53/88.47	89.74/84.67/87.13/77.20/97.05/85.47
	EFICNN	2025	94.80/94.88/94.84/90.19/98.67/94.08	96.50/90.73/93.53/87.85/98.38/92.60	94.82/85.28/89.80/81.49/97.50/88.38
	Ours	-	**95.92**/94.39/**95.15/90.75/98.86/94.51**	96.23/91.52/**93.82/88.36/98.58/93.02**	96.97/**86.78/91.71/85.56/98.19/90.81**
WHU-CD	ICIFNet	2022	89.86/85.01/87.37/77.57/99.03/86.86	87.16/85.66/86.40/76.06/98.93/85.85	80.26/86.26/83.15/71.16/98.61/82.43
	MSCANet	2022	92.20/88.17/90.14/82.05/99.24/89.74	87.35/91.21/89.24/80.57/99.13/88.78	92.75/83.26/87.75/78.17/99.08/87.27
	DMINet	2023	88.77/86.08/87.04/77.63/99.02/86.89	83.32/85.39/84.34/72.92/98.74/83.69	79.01/84.19/81.52/68.80/98.49/80.73
	AMTNet	2023	91.60/90.30/90.95/83.40/99.29/90.58	90.36/89.70/90.03/81.86/99.21/89.62	88.07/87.25/87.66/78.03/99.03/87.16
	DA-IDANet	2024	88.22/92.08/90.11/82.00/99.20/89.69	84.33/89.72/86.94/76.90/98.93/86.38	83.06/87.51/85.23/74.26/98.80/84.60
	STFF-GA	2024	85.94/89.04/87.47/77.72/98.99/86.94	82.77/90.49/86.46/76.15/98.88/85.87	81.12/**89.25**/84.99/73.90/98.75/84.34
	SEIFNet	2024	93.34/93.31/93.33/87.49/99.47/93.05	88.24/92.37/90.26/82.25/99.21/89.85	87.96/89.01/88.48/79.34/99.08/88.00
	ISDANet	2025	87.64/**94.05**/90.73/83.03/99.24/90.33	85.14/91.54/88.22/78.93/99.03/87.72	76.51/88.86/82.22/69.81/98.48/81.43
	EFICNN	2025	94.59/90.91/92.72/86.42/99.43/92.42	92.45/**92.57**/92.51/86.06/99.41/92.20	90.52/88.95/89.73/81.37/99.19/89.31
	Ours	-	**94.75**/92.80/**93.76/88.26/99.51/93.51**	**94.76**/91.26/**92.97/86.87/99.45/92.69**	**93.66**/87.49/**90.47/82.59/99.27/90.09**

7) SEIFNet: A spatiotemporal enhancement and interlevel fusion network effectively improves the ability of feature representation for changing objects.

8) ISDANet: A novel interactive and supervised dual-mode attention network improves the results of the ambiguous change area and small target omission.

9) EFICNN: An edge and flow guided iterative CNN, demonstrating superior feature extraction capabilities and CD performance.

C. Experimental Setup

1) Implementation Details: All the experiments are conducted using the PyTorch framework with a GeForce RTX 3090Ti GPU. The proposed network is optimized using AdamW optimizer for 150 epochs. The learning rate is set to 0.0001, which decays linearly with the epochs, and the batch size is set to 8. Meanwhile, the data augmentation techniques, including random flip, rotate, and crop are adopted to all comparative methods. During the training stage, only noisy labels are utilized for model optimization, while GT labels are employed for metrics computation during inference.

2) Evaluation Metrics: To quantitatively evaluate the CD performance of different methods, six widely recognized metrics are employed, including precision, recall, $F1$-score ($F1$), Intersection over Union (IoU), overall accuracy (OA), and Kappa coefficient (Kappa) [71]. The corresponding definitions are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{19}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{21}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{22}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{23}$$

$$\text{PRE} = \frac{(\text{TP} + \text{FN})(\text{TP} + \text{FP}) + (\text{TN} + \text{FN})(\text{TN} + \text{FP})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2} \tag{24}$$

$$\text{Kappa} = \frac{\text{OA} - \text{PRE}}{1 - \text{PRE}} \tag{25}$$

where TP, TN, FP, and FN represent true-positive, true-negative, false-positive, and false-negative pixels, respectively. "Negative" indicates the absence of change, while "True" signifies the correctness of the prediction.

D. Performance Comparison

1) Quantitative Evaluation: Table I elaborates the results of all the competing methods on three datasets with different noise ratios. Values highlighted in bold and underlined indicate the best and second-best results.

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For the LEVIR-CD dataset, the comparison of our proposed FuzzyCD with the baseline models demonstrates its superior performance. For instance, when the noise ratio is set to 0.5, SEIFNet exhibits impressive results with 93.66% in precision, 84.86% in recall, 89.04% in $F1$, 80.25% in IoU, 98.94% in OA, and 88.49% in Kappa, which can be attributed to its spatiotemporal difference enhancement and adaptive context fusion, effectively capturing change features. However, the proposed FuzzyCD achieves the best performance across all six metrics with 94.10% in precision, 86.68% in recall, 90.23% in $F1$, 82.21% in IoU, 99.04% in OA, and 89.73% in Kappa, outperforming the SEIFNet by 0.44%, 1.82%, 1.19%, 1.96%, 0.1%, and 1.24%, respectively. Furthermore, as reported in Table I, it can be observed that experimental results demonstrate a consistent performance degradation across all methods as the noise ratio increases. However, our model manifests the smallest decline, indicating its enhanced robustness against noisy labels. Specifically, with noise ratios ranging from 0 to 0.5 and 0.5 to 1, the IoU generated by FuzzyCD decreases 1.87%, 3.3%, while ISDANet for 1.84%, 4.93%, and EFICNN for 3.42%, 4.83%, respectively.

For the CDD dataset, compared to other competing methods, the SEIFNet and EFICNN achieve competitive performance, ranking third and second, respectively, only surpassed by our proposed FuzzyCD. When the noise ratio is set to 1, as indicated in Table I, the FuzzyCD outperforms SEIFNet and EFICNN on five metrics. Although the Precision of FuzzyCD is lower than the two methods, indicating a higher number of false positives, it achieves the highest recall, $F1$, IoU, OA, and Kappa, which, respectively, surpass those of SEIFNet by 5.2%, 2.85%, 5.6%, 0.6%, and 3.29%, and those of EFICNN by 1.5%, 1.91%, 4.07%, 0.69%, and 2.43%. Similar to the observation on LEVIR-CD dataset, FuzzyCD exhibits the smallest reduction in IoU among all compared methods. From noise ratio 0 to 1, the IoU of FuzzyCD decreases 5.19%, while STFF-GA for 5.28%, and MSCANet for 6.96%.

As for the WHU-CD dataset, the FuzzyCD exhibits a relative improvement across different evaluation metrics, achieving the best or second-best quantitative values. Under the noise ratio of 0.5, AMTNet that employs attention and transformer modules to model contextual information attains 90.36% in precision, 89.70% in recall, 90.03% in $F1$, 81.86% in IoU, 99.21% in OA, and 89.62% in Kappa. It can be observed that the quantitative results of ISDANet exhibit a more pronounced decline in IoU, with specific values of 13.22% from noise ratio 0 to 1, while MSCANet, STFF-GA, and FuzzyCD decrease 3.88%, 3.82%, and 5.67%, respectively. Although FuzzyCD exhibits a relatively greater performance decline compared to MSCANet and STFF-GA, it maintains superior metric values across all evaluation criteria.

To further demonstrate how the quantitative results evolve with increasing noise ratios, Fig. 6 presents a radar chart visualization across evaluation metrics. As clearly illustrated, under noise-free conditions, the performance of all competing methods remains closely clustered, with the results of SEIFNet and EFICNN almost overlapping with those of FuzzyCD. However, as the noise ratio increases, FuzzyCD significantly outperforms other competing methods, quantitatively demonstrating its improved performance in the presence of noisy labels.

2) Qualitative Evaluation: The visualization results of each competing method on three datasets are shown in Figs. 7–9. To provide a more intuitive demonstration of the performance across different methods, in these visualization results, TP, TN, FN, and FP are marked in white, black, red, and green, respectively.

Fig. 7 presents the visualization results on LEVIR-CD dataset. As shown in the figure, under noise-free conditions (i.e., $\rho = 0$), comparative methods can generate satisfactory CD results. However, when the noise ratio is set to 0.5, all comparison methods exhibit varying degrees of missed detections, particularly at edge regions. In contrast, the proposed FuzzyCD achieves better identification for edge details of changes and maintains the overall integrity of change regions, fully demonstrating its advantages in dealing with noisy labels, which is significant in practice. This phenomenon becomes more pronounced when the noise ratio reaches 1; these methods struggle to accurately locate change regions. Taking (h) as an example, due to the interference arising from noisy labels, the competing methods erroneously classify the changed pixels at building boundaries as unchanged, whereas the change map generated by FuzzyCD has been significantly improved, generating considerably fewer false negatives. This improvement can be attributed to the incorporation of fuzzy logic, which effectively enhances the capability of the proposed model to capture and represent uncertain information, thereby improving the model's robustness against noisy labels.

The visualization results on CDD dataset are presented in Fig. 8. As shown in the figure, when the noise ratio is set to 0, baseline methods tend to generate false positives, while FuzzyCD effectively mitigates pseudo-changes. Taking sample Fig. 8(c) as an example, most competing methods erroneously identify the unchanged areas between the change objects as changed, causing the originally discrete objects to merge as a single entity. Only SEIFNet, EFICNN, and FuzzyCD are capable of outputting satisfactory CD results. Under the scenario depicted in Fig. 8(d), both ISDANet and FuzzyCD can still detect every single road, maintaining high identification and low false alarms. Notably, the qualitative analysis of the visualization results indicates that all methods struggle with detecting small objects due to the negative impact of noisy labels, as evidenced in Fig. 8(c) and (g). Under a noise ratio of 1, the competing methods exhibit extensive erroneous detection areas, suggesting their sensitivity to noisy labels. Compared to them, the proposed FuzzyCD achieves more accurate CD results.

Qualitative results on WHU-CD dataset reveal the relative improvements of FuzzyCD over other competing methods. As shown in Fig. 9, under noise-free scenarios, all comparative methods are able to locate genuine change regions with minimal false alarms. As the noise ratio increases to 0.5, previous methods begin to suffer from significant noise interference. As illustrated in Fig. 9(d), ICIFNet, DA-IDANet, and STFF-GA misidentify the bottom portion of the change targets as unchanged, while AMTNet, SEIFNet, and ISDANet

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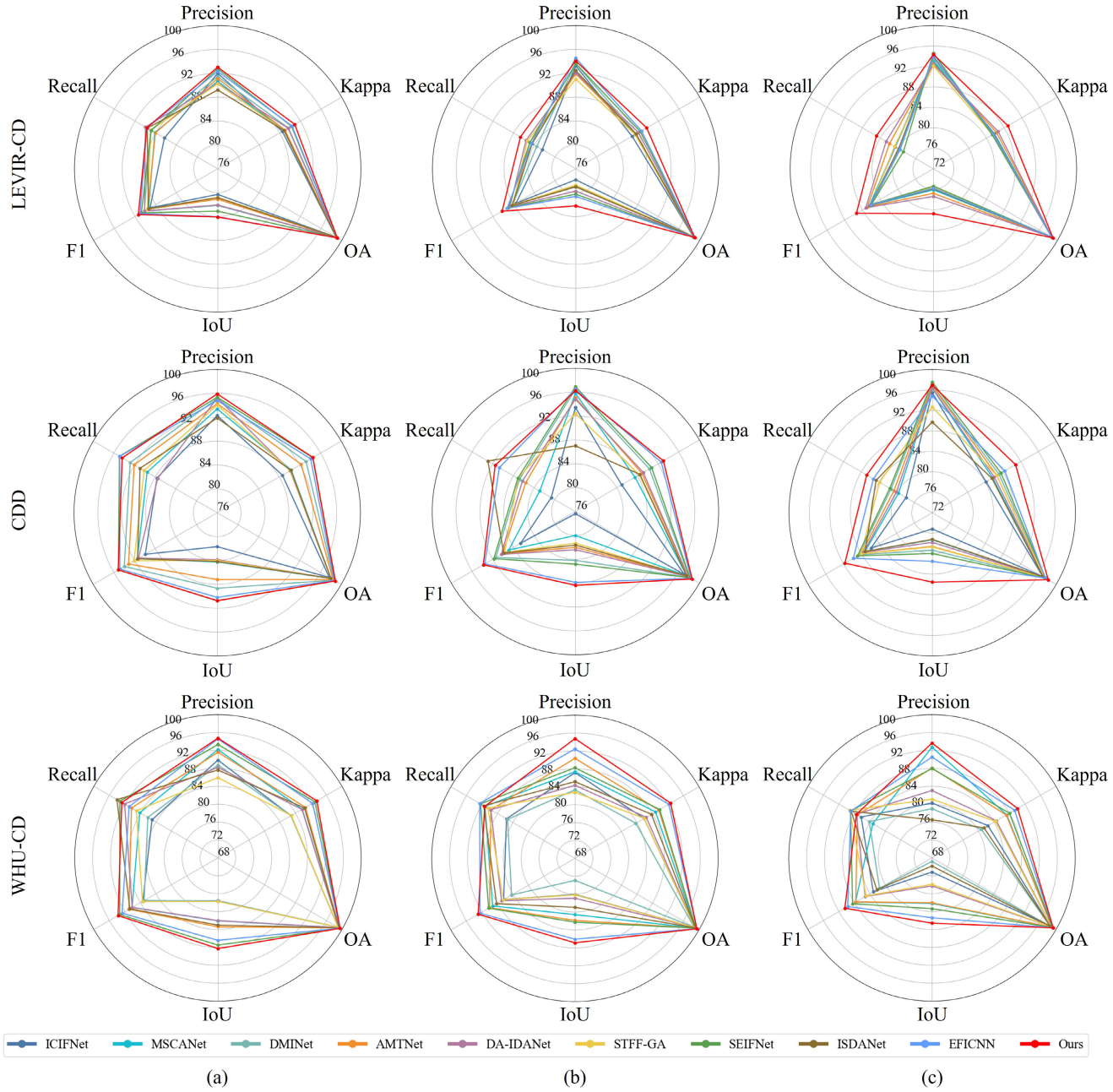


Fig. 6. Evaluation metric radar chart. The above figures demonstrate the evaluation metric radar charts on LEVIR-CD, CDD, and WHU-CD. Each column represents a different noise ratio. Specifically, (a)–(c) correspond to the noise ratios $\rho = 0$, $\rho = 0.5$, and $\rho = 1$, respectively.

erroneously classify the right-side unchanged areas as changed. In contrast, EFICNN and the proposed FuzzyCD demonstrate enhanced robustness, relatively reducing the erroneous identification areas. Similar patterns emerge in Fig. 9(f), where only MSCANet, AMTNet, EFICNN, and FuzzyCD maintain complete CD results, while other methods fail to detect the upper change regions. When dealing with small targets as shown in Fig. 9(h), FuzzyCD achieves comprehensive detection while maintaining the integrity of external edges. At maximum noise (i.e., $\rho = 1$), the comparison methods may misclassify the building targets at the bottom as background due to their highly similar colors, which are particularly deceptive. However, FuzzyCD significantly improves the quality of visualization results, demonstrating its superior performance against noisy labels.

3) Efficiency Evaluation: In addition to the quantitative and qualitative evaluation previously discussed, efficiency assessment is equally important for comprehensive model evaluation. To thoroughly analyze the computational efficiency of our proposed FuzzyCD, the number of parameters (Params.), floating-point operations (FLOPs), inference time with input image sizes of 256×256, and training time on LEVIR-CD dataset for all methods are presented in Table II.

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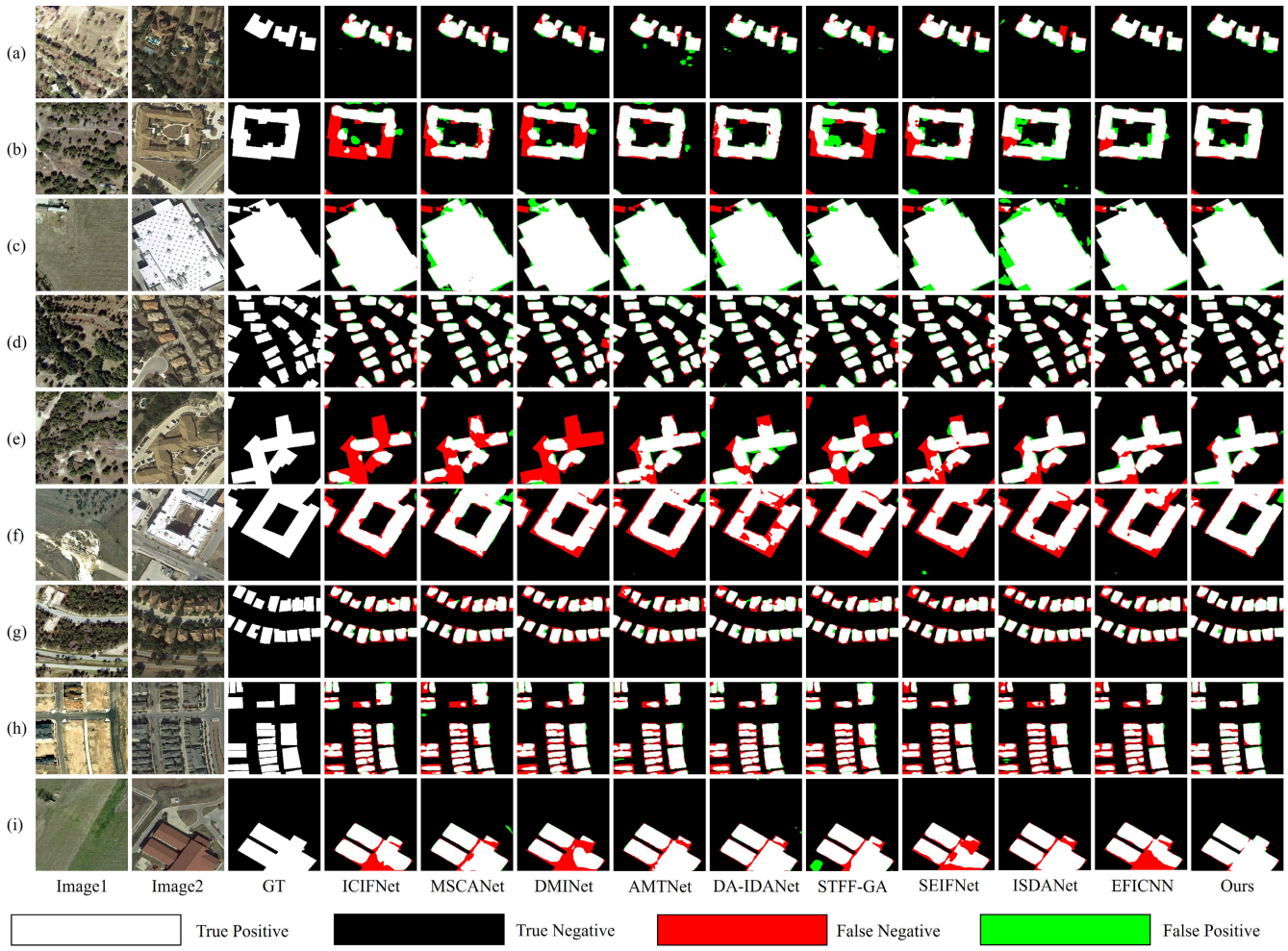


Fig. 7. Visualization results on LEVIR-CD dataset. (a)–(c) Correspond to the noise ratio of 0, while (d)–(f) correspond to the noise ratio of 0.5, and (g)–(i) correspond to the noise ratio of 1.

TABLE II
EFFICIENCY EVALUATION FOR ALL COMPARED METHODS

Methods	Params. (M)	FLOPs (G)	Inference time (s)	Training time (h)
ICIFNet	23.843	25.410	0.017	11.667
MSCANet	16.421	14.802	0.011	3.055
DMINet	6.242	14.551	0.005	7.223
AMTNet	24.665	21.559	0.011	4.305
DA-IDANet	16.416	49.346	0.011	6.945
STFF-GA	22.365	11.668	0.012	5.625
SEIFNet	27.905	8.371	0.005	3.610
ISDANet	6.404	3.796	0.015	6.250
EFICNN	21.325	91.340	0.016	5.417
Ours	24.993	26.518	0.009	5.125

From the results, it can be observed that ISDANet achieves optimal computational efficiency due to its lightweight design using MobileNetV2 as the backbone. While our proposed method exhibits marginally higher computational demands compared to other methods, this negligible sacrifice is justified by its improved CD performance and enhanced robustness to noisy labels, a worthwhile tradeoff given contemporary computational capabilities.

E. Ablation Experiments

In this section, to independently evaluate the effectiveness of each component of our proposed method, comprehensive ablation experiments are conducted across three datasets. For these experiments, identical training strategies are employed for all models, and individual components are systematically integrated into the backbone network to specify their respective contributions.

1) Number of Fuzzy Units: To illustrate the impact of the number of fuzzy units (i.e., EUEU and AUEU) on model performance, experiments are conducted by varying the number of units from 1 to 4. As reported in Table III, FuzzyCD achieves the best results when configured with 3 units, which can be attributed to its effective balance between model complexity and capacity. Specifically, excessive complexity may lead to overfitting on noisy labels, while insufficient complexity fails to capture underlying data patterns. Experimental results confirm this observation, indicating performance degradation at both 2 and 4 units. Therefore, the number of EUEU and AUEU is configured to 3 in this article.

2) Effectiveness of EUEU and AUEU: As acknowledged, fuzzy neural networks excel at processing uncertain and

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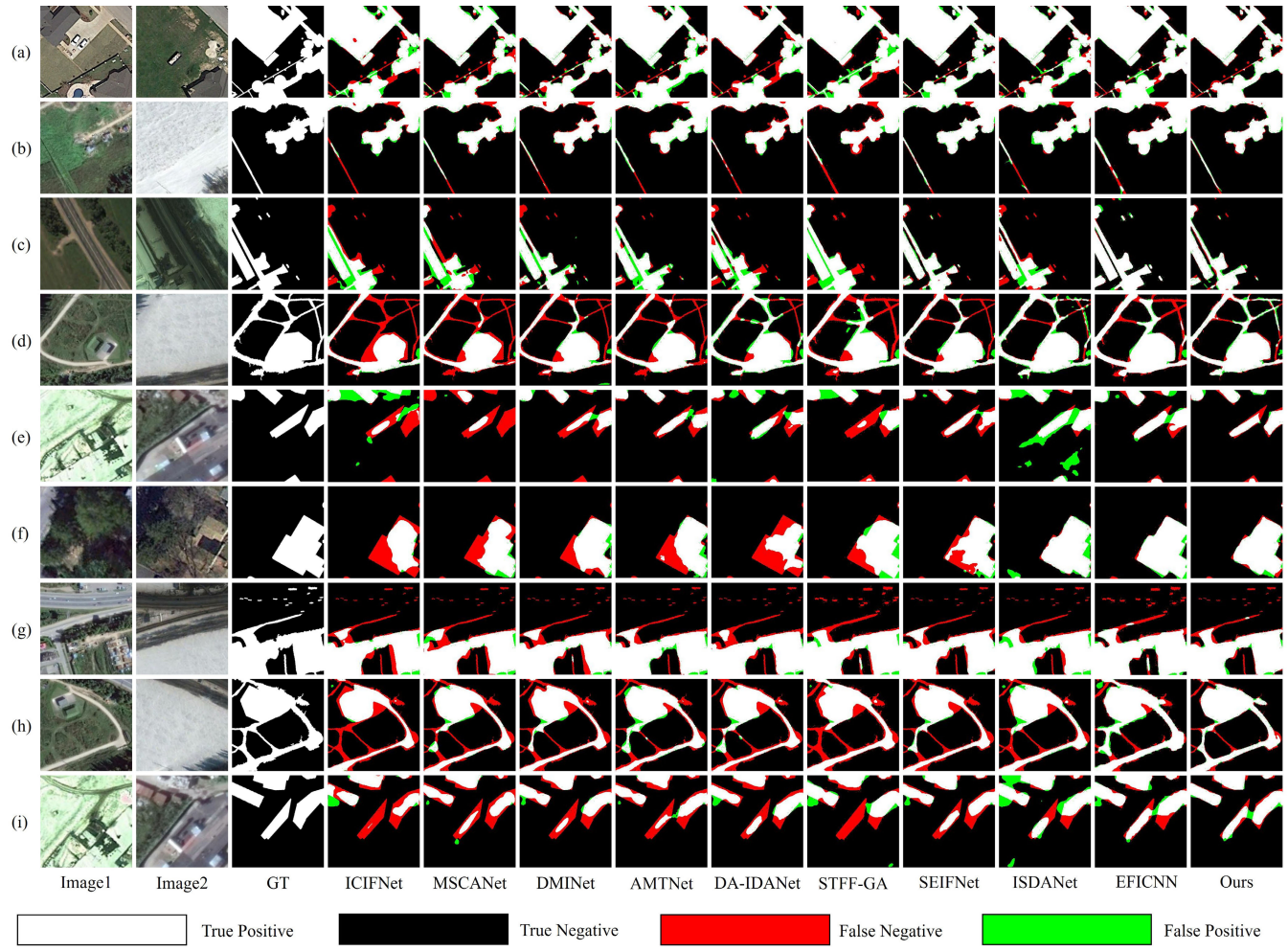


Fig. 8. Visualization results on CDD dataset. (a)–(c) Correspond to the noise ratio of 0, while (d)–(f) correspond to the noise ratio of 0.5, and (g)–(i) correspond to the noise ratio of 1.

TABLE III
ABLATION RESULTS OF NUMBER OF FUZZY UNITS(%)

Datasets	Number of units	Precision	Recall	F1	IoU	OA	Kappa
LEVIR-CD	1	90.03	87.79	88.90	80.01	98.88	88.31
	2	92.71	89.02	90.83	83.20	99.08	90.35
	3 (Ours)	93.03	**89.73**	**91.35**	**84.08**	**99.13**	**90.90**
	4	**93.05**	89.52	91.25	83.91	**99.13**	90.79
CDD	1	94.29	91.60	92.93	86.79	98.35	91.99
	2	**97.12**	89.48	93.14	87.16	98.45	92.27
	3 (Ours)	95.92	**94.39**	**95.15**	**90.75**	**98.86**	**94.51**
	4	94.95	93.06	93.99	88.67	98.60	93.20
WHU-CD	1	90.93	90.12	90.52	82.69	99.25	90.13
	2	93.86	**92.87**	93.36	87.55	99.48	93.09
	3 (Ours)	**94.75**	92.80	**93.76**	**88.26**	**99.51**	**93.51**
	4	94.27	90.51	92.35	85.79	99.41	92.04

TABLE IV
ABLATION RESULTS OF FUZZY LOGIC UNITS(%)

Datasets	EUEU	AUEU	Precision	Recall	F1	IoU	OA	Kappa
LEVIR-CD	×	×	92.53	87.98	90.20	82.14	99.03	89.69
	√	×	92.63	89.78	91.18	83.79	99.12	90.72
	×	√	92.10	**89.92**	90.99	83.48	99.09	90.52
	√	√	**93.03**	89.73	**91.35**	**84.08**	**99.13**	**90.90**
CDD	×	×	93.59	89.72	91.62	84.53	98.06	90.52
	√	×	94.93	94.18	94.56	89.67	98.72	93.83
	×	√	92.04	92.52	92.28	87.18	98.40	92.23
	√	√	**95.92**	**94.39**	**95.15**	**90.75**	**98.86**	**94.51**
WHU-CD	×	×	85.96	**93.71**	89.66	81.26	99.14	89.22
	√	×	93.40	93.41	93.41	87.63	99.48	93.13
	×	√	90.98	91.07	91.02	83.52	99.29	90.65
	√	√	**94.75**	92.80	**93.76**	**88.26**	**99.51**	**93.51**

ambiguous information. The proposed EUEU and AUEU harness this capability through fuzzy logic integration, simultaneously enhancing CD performance and noise robustness. To validate the effectiveness of the two fuzzy logic units, relevant ablation experiments are designed, with quantitative results presented in Table IV. Taking CDD dataset as an example, it is noted that when EUEU and AUEU are removed, the CD performance demonstrates a significant degradation, with 93.59% in precision, 89.72% in recall, 91.62% in $F1$, 84.53% in IoU, 98.06% in OA, and 90.52% in Kappa. However, when EUEU and AUEU are individually incorporated, the quantitative metrics exhibit varying degrees of improvement. Specifically, EUEU improves precision, recall, $F1$, IoU, OA, and Kappa by 1.34%, 4.46%, 2.94%, 5.14%, 0.66%, and

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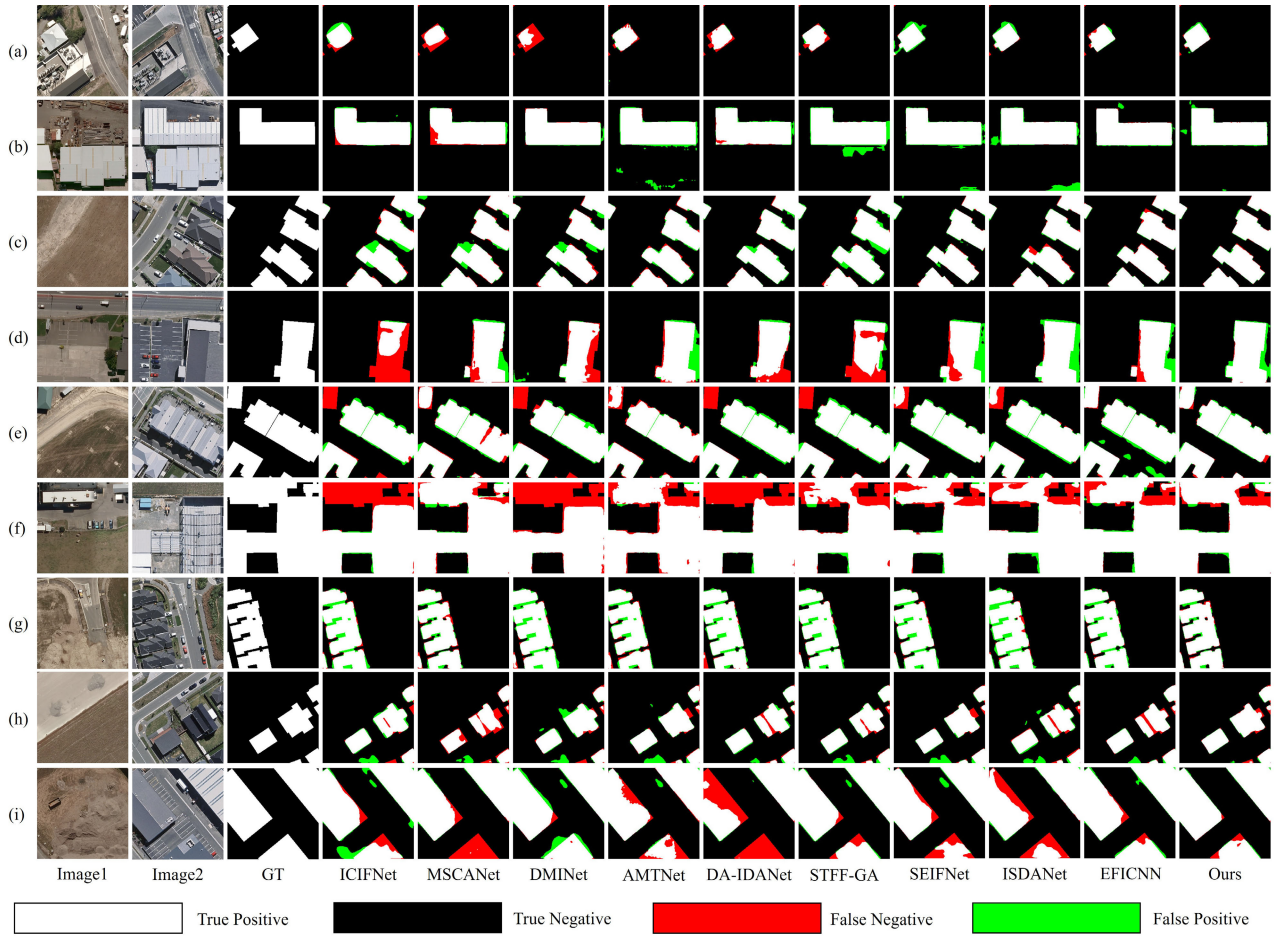


Fig. 9. Visualization results on WHU-CD dataset. (a)–(c) Correspond to the noise ratio of 0, while (d)–(f) correspond to the noise ratio of 0.5, and (g)–(i) correspond to the noise ratio of 1.

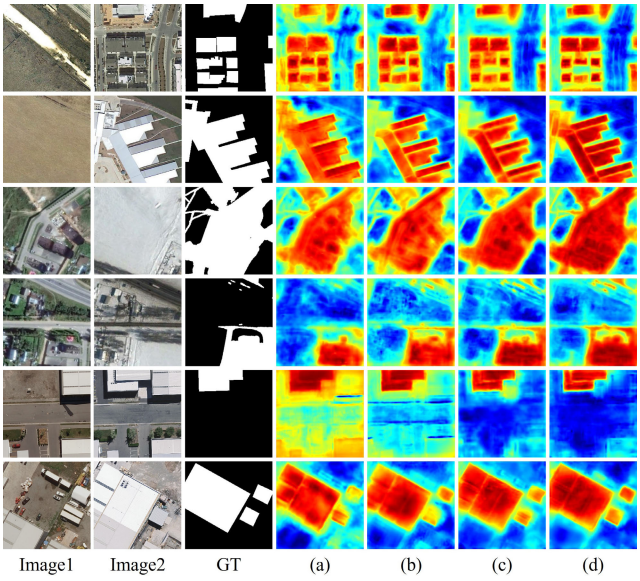


Fig. 10. Visualization of feature maps of different models. (a) Model without EUEU and AUEU. (b) Model with EUEU only (without AUEU). (c) Model with AUEU only (without EUEU). (d) Model with EUEU and AUEU. The first and second rows show samples from LEVIR-CD, the third and fourth from CDD, and the fifth and sixth from WHU-CD.

3.31%, respectively, while AUEU achieves gains of 2.8%, 0.66%, 2.65%, 0.34%, and 1.71% in recall, $F1$, IoU, OA, and Kappa, respectively. FuzzyCD achieves superior performance when both EUEU and AUEU are integrated, with consistent results observed on LEVIR-CD and WHU-CD datasets. This can be attributed to that the fuzzy logic units pay more attention to uncertainties induced by noisy labels, thereby effectively mitigating the negative impact of label noise.

To provide a more intuitive demonstration of the proposed fuzzy logic units, the feature maps of several representative samples from three datasets are visualized in Fig. 10. The visualization results reveal that the removal of EUEU and AUEU impairs the CD performance, as it fails to extract uncertain information to deal with the noisy labels. Conversely, FuzzyCD with EUEU and AUEU demonstrates refined detection for the edge structure of change regions, effectively improving the quality of CD results.

3) Discussion of Weighting Factor λ: The loss weights serve to modulate the relative contributions of individual loss terms to the overall optimization objective, thereby enabling the control of their hierarchical importance. In this article, the weighting factor λ is introduced to balance the interplay between negative log-likelihood and CE losses, steering the model toward optimal learning behavior. To investigate the best value of λ, a systematic ablation study is conducted by varying λ from 0 to 1 with the following three metrics: $F1$, IoU, and Kappa. The results are illustrated in Fig. 11,

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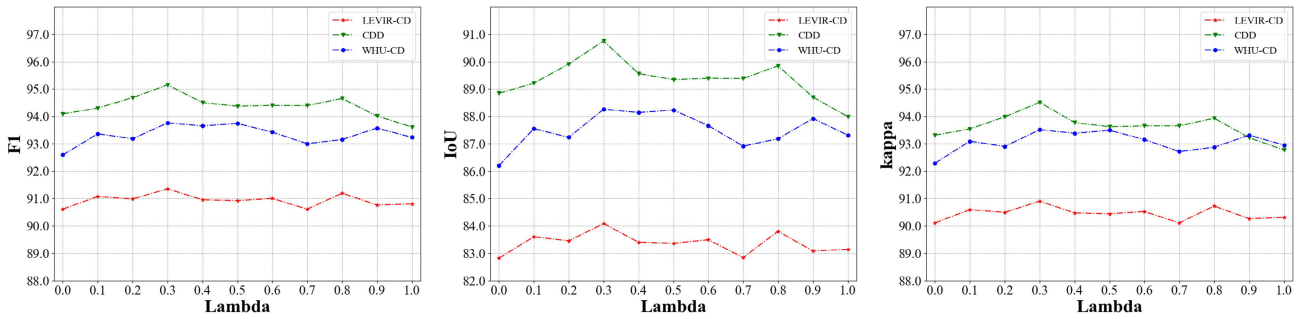


Fig. 11. Effects of varying weighting factor λ from 0 to 1 on evaluation metrics: (a) $F1$, (b) IoU, and (c) Kappa.

without incorporating negative log-likelihood loss (i.e., $\lambda = 0$), leading to the poorest performance. Notably, the configuration with $\lambda = 0.3$ achieves peak performance. This optimal balance reflects the synergistic effect of probabilistic uncertainty modeling with negative log-likelihood and discriminative classification with CE.

IV. Conclusion

In conclusion, a novel fuzzy logic-integrated network is proposed for robust CD with noisy labels, termed FuzzyCD. It successfully mitigates the interference from noisy labels through the introduction of fuzzy logic. Specifically, an EUEU is proposed to effectively model the epistemic uncertainty arising from noisy labels. This is achieved by employing XOR operation as fuzzy logic for feature fuzzification, thereby alleviating the impact of epistemic uncertainty and improving the CD performance. Besides, an AUEU is designed to capture inherent data corruption, which characterizes aleatoric uncertainty through probabilistic inference. By integrating fuzzy logic into the CD model, the proposed FuzzyCD achieves improved CD performance as well as enhanced model robustness against noisy labels. Extensive experiments and comparisons demonstrate that our proposed model outperforms existing methods on three CD datasets. However, our work is built upon the CNN framework, leaving the recently popular Mamba and diffusion frameworks unexplored. Whether this limitation would restrict the generalization ability of our proposed method requires further investigation. For future work, we intend to explore diverse framework architectures to better understand the applicability of the proposed method. Additionally, we plan to validate the effectiveness of our proposed method on other CD tasks, such as heterogeneous CD and nonregistered CD scenarios.

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